

DATA SCIENCE PROGRAMMING
AND TOOLS

CUSTOMER SHOPPING TRENDS DATASET

By 68056034 ณัฐชิวา มาลากร



CONTEXT

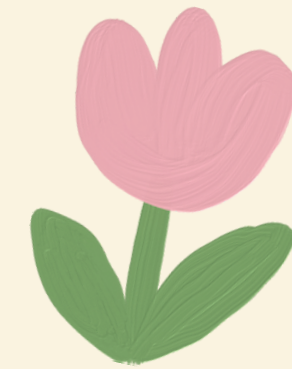


The Consumer Behavior and Shopping Habits Dataset provides comprehensive insights into consumer preferences, trends, and patterns during their shopping experiences. This dataset encompasses a wide variety of variables, including demographic information, purchase history, product preferences, and purchase frequency. With this complete dataset, analysts and researchers can delve into the complexities of the consumer decision-making process.





Overview

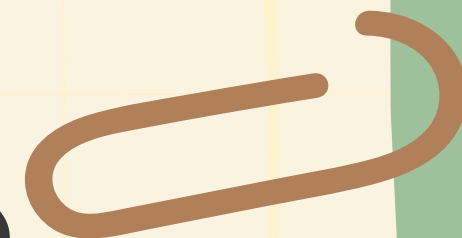


- 1 **IMPORT & LOAD DATA**
- 2 **DATA CLEANING**
- 3 **EXPLORATORY DATA ANALYSIS (EDA)**
- 4 **BUSINESS INSIGHTS & KEY FINDINGS**
- 5 **USER INTERFACE (UI)**





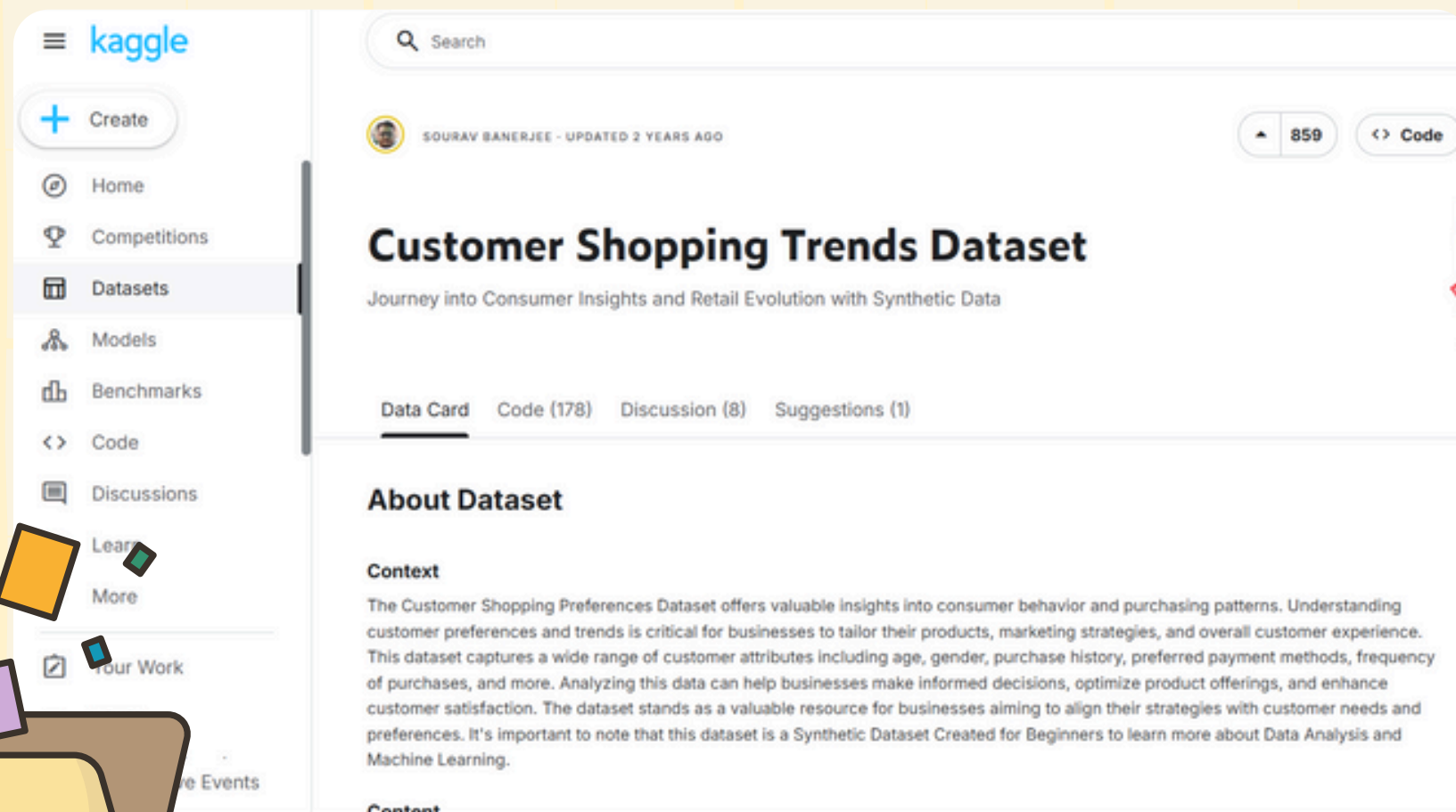
IMPORT & LOAD DATA



```
import kagglehub

# Download latest version
path = kagglehub.dataset_download("zeesolver/consumer-behavior-and-shopping-habits-dataset")

print("Path to dataset files:", path)
```



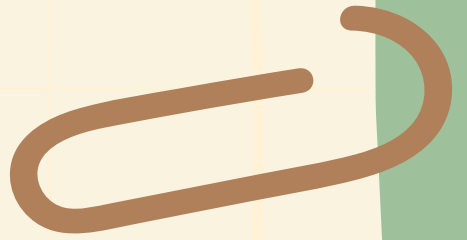
```
#Basic libraries
import pandas as pd
import numpy as np

#Libraries for visualization
import seaborn as sns
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')
```





IMPORT & LOAD DATA



- ข้อมูลเบื้องต้นของตาราง shopping_behavior

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually



DATA CLEANING



Data Types

- `.info()`
- `.shape()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                      Non-Null Count  Dtype
---  -
0   Customer ID                 3900 non-null   int64
1   Age                         3900 non-null   int64
2   Gender                       3900 non-null   object
3   Item Purchased              3900 non-null   object
4   Category                    3900 non-null   object
5   Purchase Amount (USD)       3900 non-null   int64
6   Location                    3900 non-null   object
7   Size                        3900 non-null   object
8   Color                       3900 non-null   object
9   Season                      3900 non-null   object
10  Review Rating                3900 non-null   float64
11  Subscription Status          3900 non-null   object
12  Shipping Type                3900 non-null   object
13  Discount Applied             3900 non-null   object
14  Promo Code Used              3900 non-null   object
15  Previous Purchases           3900 non-null   int64
16  Payment Method               3900 non-null   object
17  Frequency of Purchases       3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

Rows, cols: (3900, 18)

DATA CLEANING



Missing values

- `.isnull().sum()`

Duplicates()

```
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD) 0
Location         0
Size             0
Color            0
Season           0
Review Rating    0
Subscription Status 0
Shipping Type    0
Discount Applied 0
Promo Code Used  0
Previous Purchases 0
Payment Method   0
Frequency of Purchases 0
dtype: int64
```

```
Orders Duplicates: 0
```

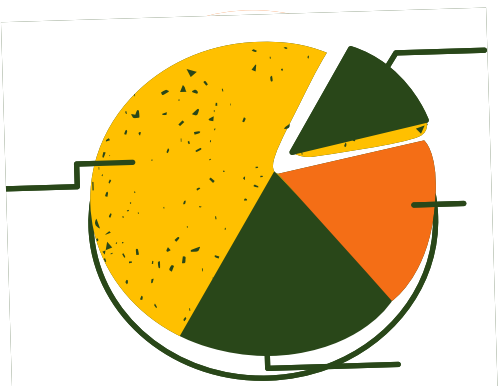
EXPLORATORY DATA ANALYSIS (EDA)

- Check columns
- Describe

```
Index(['Customer ID', 'Age', 'Gender', 'Item Purchased', 'Category',  
      'Purchase Amount (USD)', 'Location', 'Size', 'Color', 'Season',  
      'Review Rating', 'Subscription Status', 'Shipping Type',  
      'Discount Applied', 'Promo Code Used', 'Previous Purchases',  
      'Payment Method', 'Frequency of Purchases'],  
      dtype='object')
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating \
count	3900.000000	3900.000000	3900.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.749949
std	1125.977353	15.207589	23.685392	0.716223
min	1.000000	18.000000	20.000000	2.500000
25%	975.750000	31.000000	39.000000	3.100000
50%	1950.500000	44.000000	60.000000	3.700000
75%	2925.250000	57.000000	81.000000	4.400000
max	3900.000000	70.000000	100.000000	5.000000

	Previous Purchases
count	3900.000000
mean	25.351538
std	14.447125
min	1.000000
25%	13.000000
50%	25.000000
75%	38.000000
max	50.000000



EXPLORATORY DATA ANALYSIS (EDA)



Distribution

- Category
- Item Purchased
- Location
- Frequency of Purchases
- Season

```
array(['Clothing', 'Footwear', 'Outerwear', 'Accessories'], dtype=object)
```

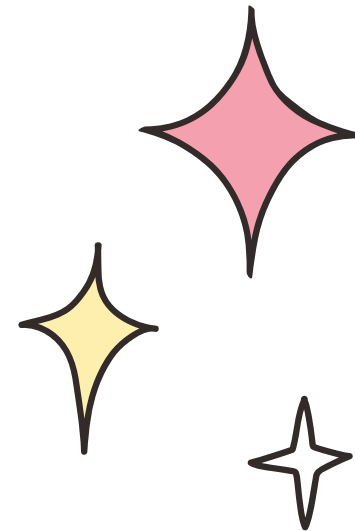
```
array(['Blouse', 'Sweater', 'Jeans', 'Sandals', 'Sneakers', 'Shirt',  
      'Shorts', 'Coat', 'Handbag', 'Shoes', 'Dress', 'Skirt',  
      'Sunglasses', 'Pants', 'Jacket', 'Hoodie', 'Jewelry', 'T-shirt',  
      'Scarf', 'Hat', 'Socks', 'Backpack', 'Belt', 'Boots', 'Gloves'],  
      dtype=object)
```

```
array(['Kentucky', 'Maine', 'Massachusetts', 'Rhode Island', 'Oregon',  
      'Wyoming', 'Montana', 'Louisiana', 'West Virginia', 'Missouri',  
      'Arkansas', 'Hawaii', 'Delaware', 'New Hampshire', 'New York',  
      'Alabama', 'Mississippi', 'North Carolina', 'California',  
      'Oklahoma', 'Florida', 'Texas', 'Nevada', 'Kansas', 'Colorado',  
      'North Dakota', 'Illinois', 'Indiana', 'Arizona', 'Alaska',  
      'Tennessee', 'Ohio', 'New Jersey', 'Maryland', 'Vermont',  
      'New Mexico', 'South Carolina', 'Idaho', 'Pennsylvania',  
      'Connecticut', 'Utah', 'Virginia', 'Georgia', 'Nebraska', 'Iowa',  
      'South Dakota', 'Minnesota', 'Washington', 'Wisconsin', 'Michigan'],  
      dtype=object)
```

```
array(['Fortnightly', 'Weekly', 'Annually', 'Quarterly', 'Bi-Weekly',  
      'Monthly', 'Every 3 Months'], dtype=object)
```

```
array(['Winter', 'Spring', 'Summer', 'Fall'], dtype=object)
```

BUSINESS INSIGHTS & KEY FINDINGS

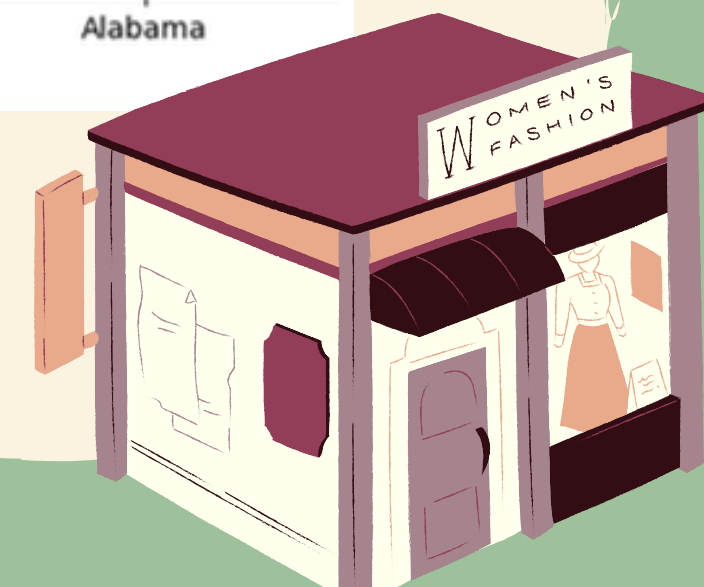


1. Top 5 Customer Locations



	Location	Number of Customers
25	Montana	96
4	California	95
11	Idaho	93
12	Illinois	92
0	Alabama	89

- **Finding:** Most customers are in Montana, California, Idaho, Illinois, and Alabama, with similar numbers.
- **Insight:** Customer distribution is fairly even.

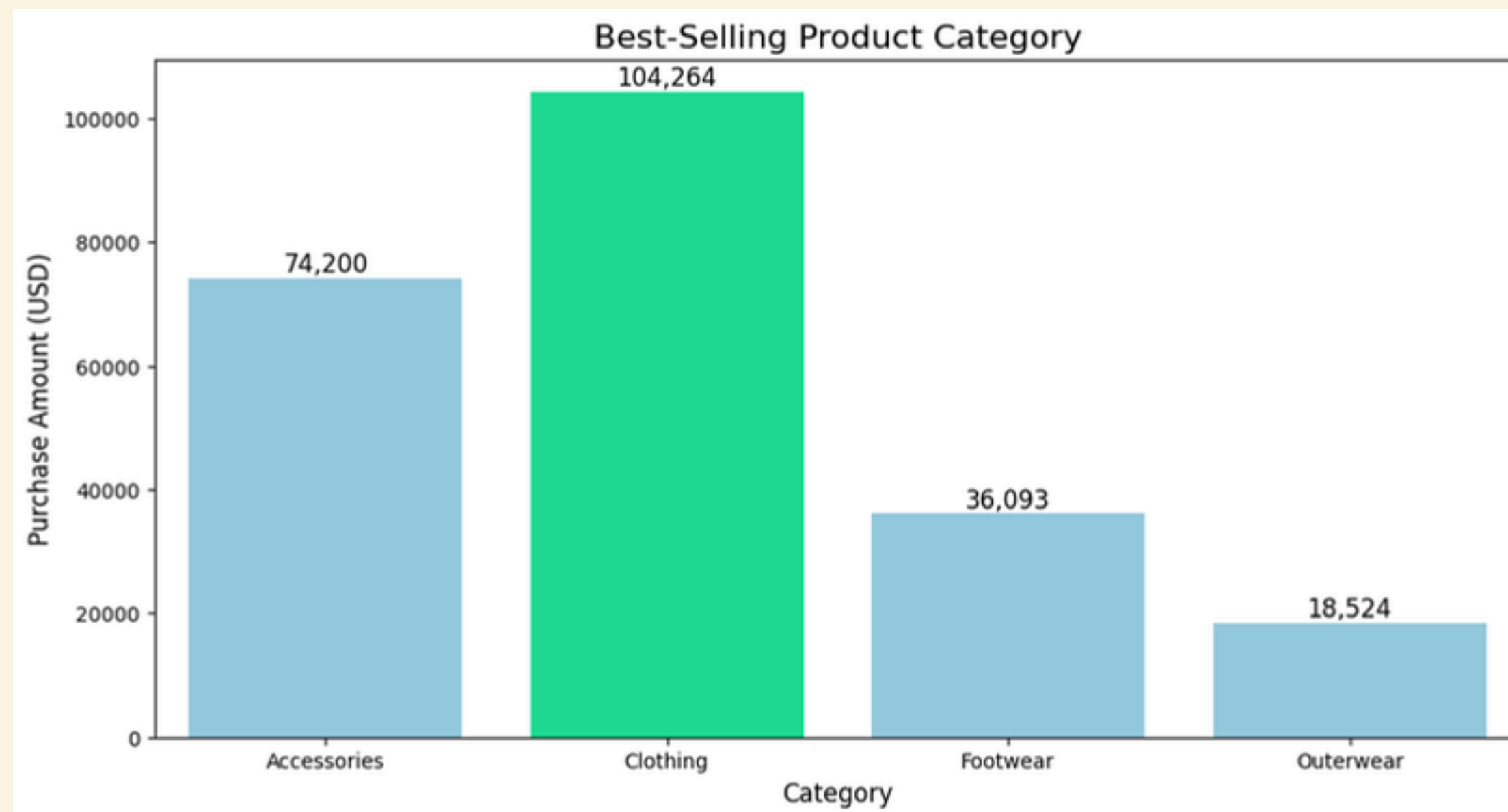




2. Total Purchase by Category

	Category	Purchase Amount (USD)
0	Accessories	74200
1	Clothing	104264
2	Footwear	36093
3	Outerwear	18524

- **Finding:** Clothing has the highest sales, followed by Accessories, Footwear, and Outerwear.
- **Insight:** Core products are Clothing and Accessories.

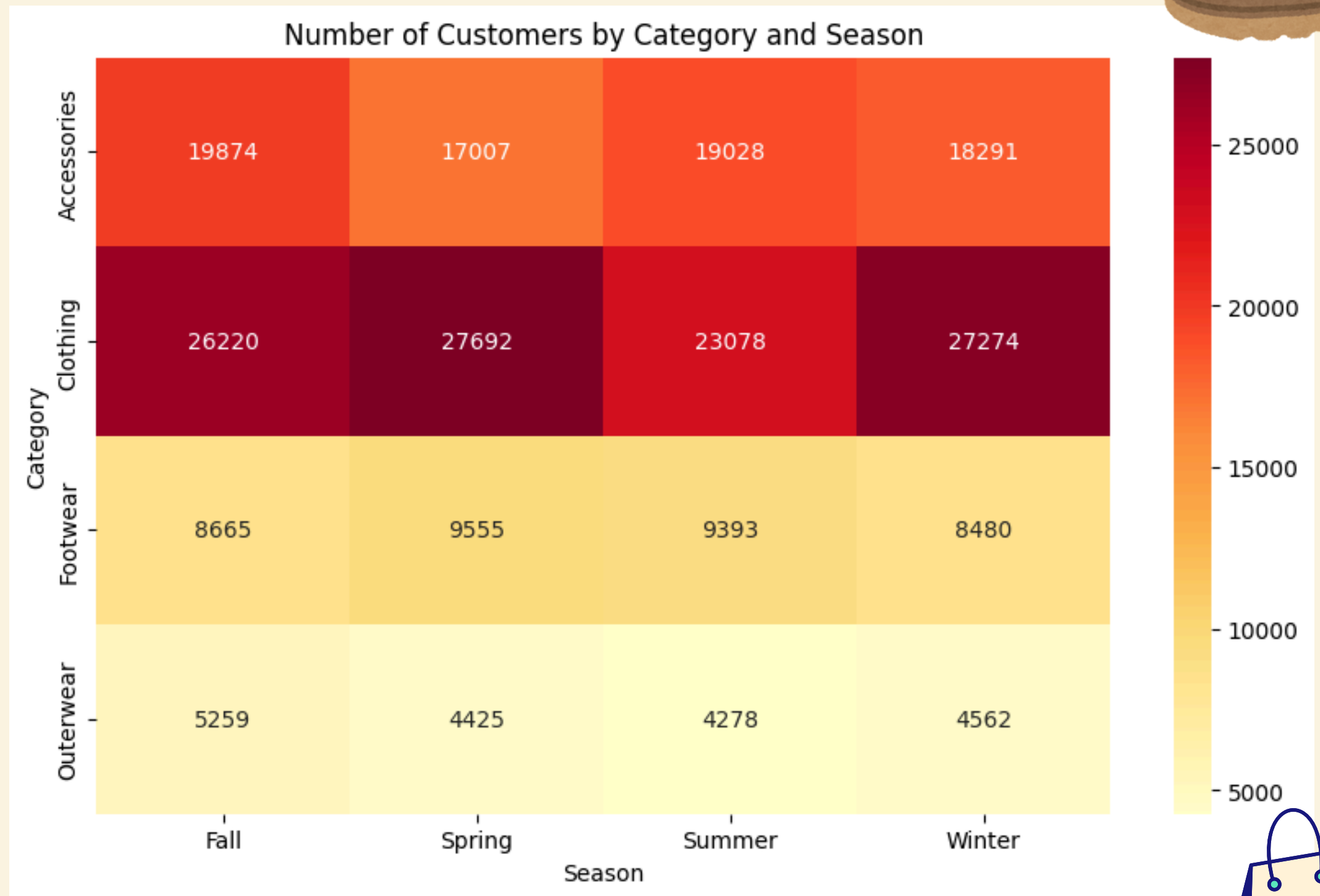


3. Sales by Season & Category



Season	Fall	Spring	Summer	Winter
Category				
Accessories	19874	17007	19028	18291
Clothing	26220	27692	23078	27274
Footwear	8665	9555	9393	8480
Outerwear	5259	4425	4278	4562

- **Finding:** Clothing has high sales across all seasons, while Accessories peak in Summer and Fall.
- **Insight:** Clothing is the core product, Accessories are seasonal products.

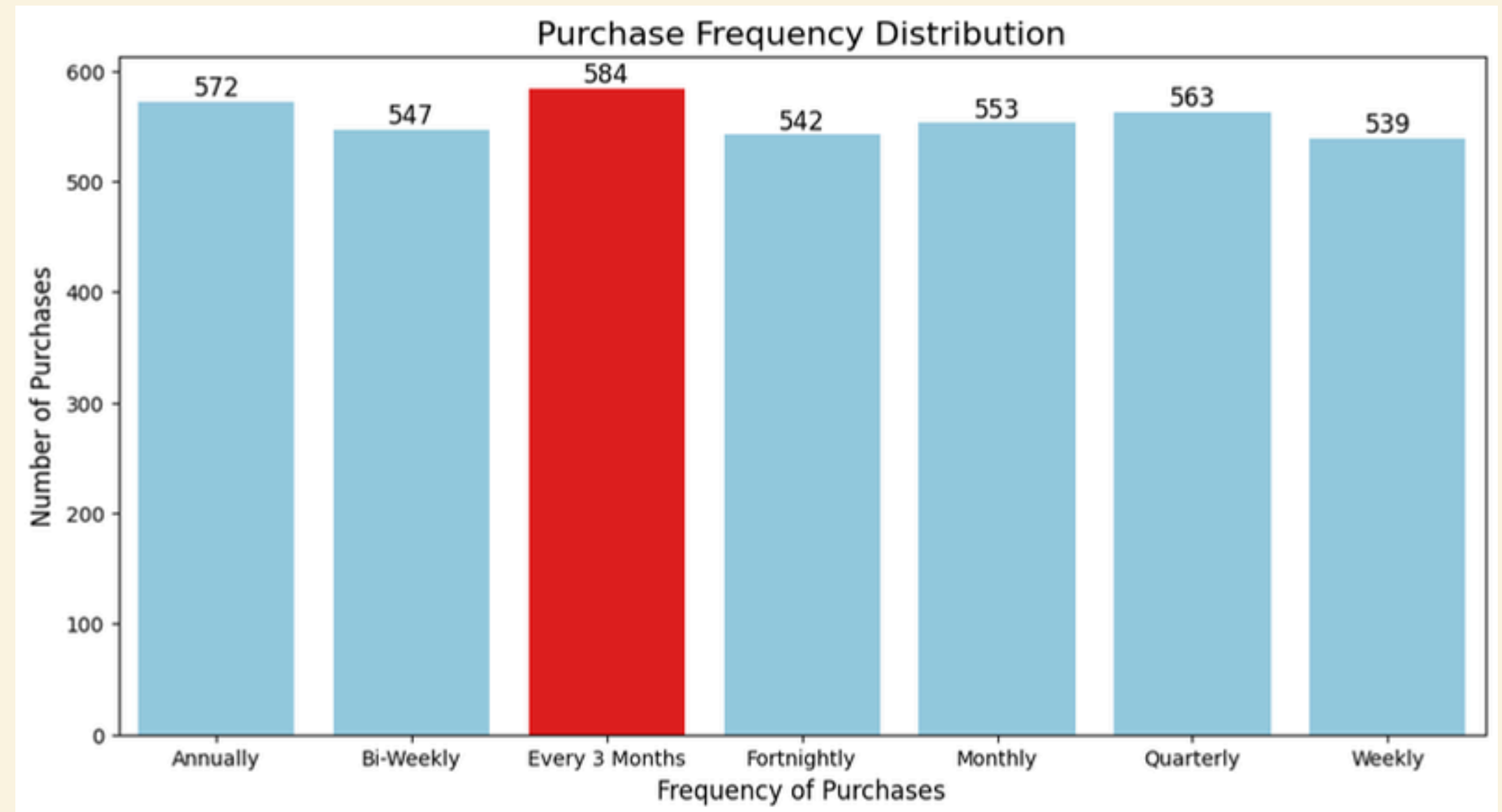


4. Customer Purchase Frequency Analysis



	Frequency of Purchases	Number of Customers
0	Annually	572
1	Bi-Weekly	547
2	Every 3 Months	584
3	Fortnightly	542
4	Monthly	553
5	Quarterly	563
6	Weekly	539

- **Finding:** Most customers purchase every 3 months or annually.
- **Insight:** Customer engagement is low; they do not buy frequently.



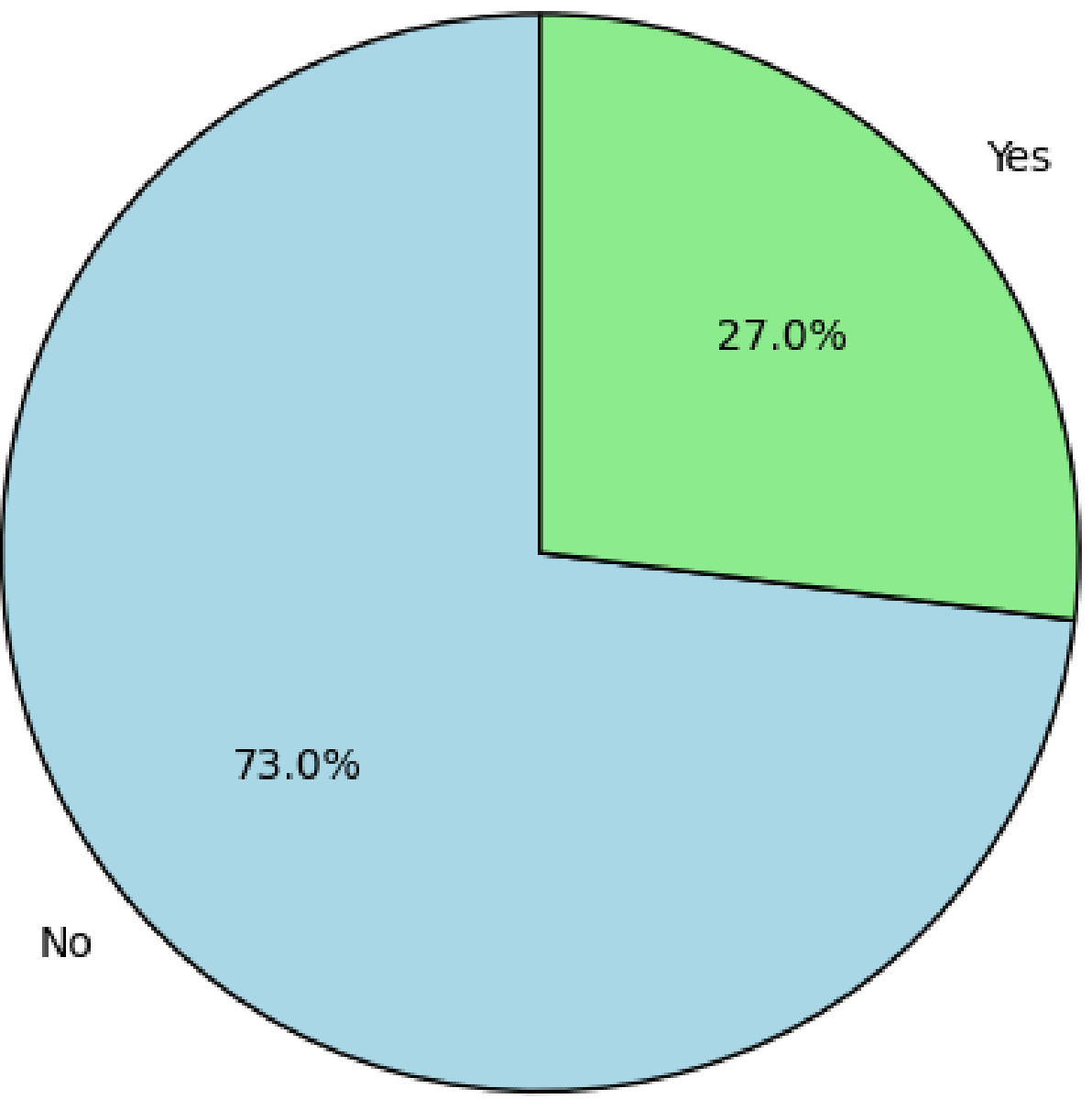


5. Membership Status

Number of Customers	
Subscription Status	
No	2847
Yes	1053

- **Finding:** 73% Non-Subscribers, 27% Subscribers
- **Insight:** Most customers are not yet subscribed.

Customer Distribution by Subscription Status

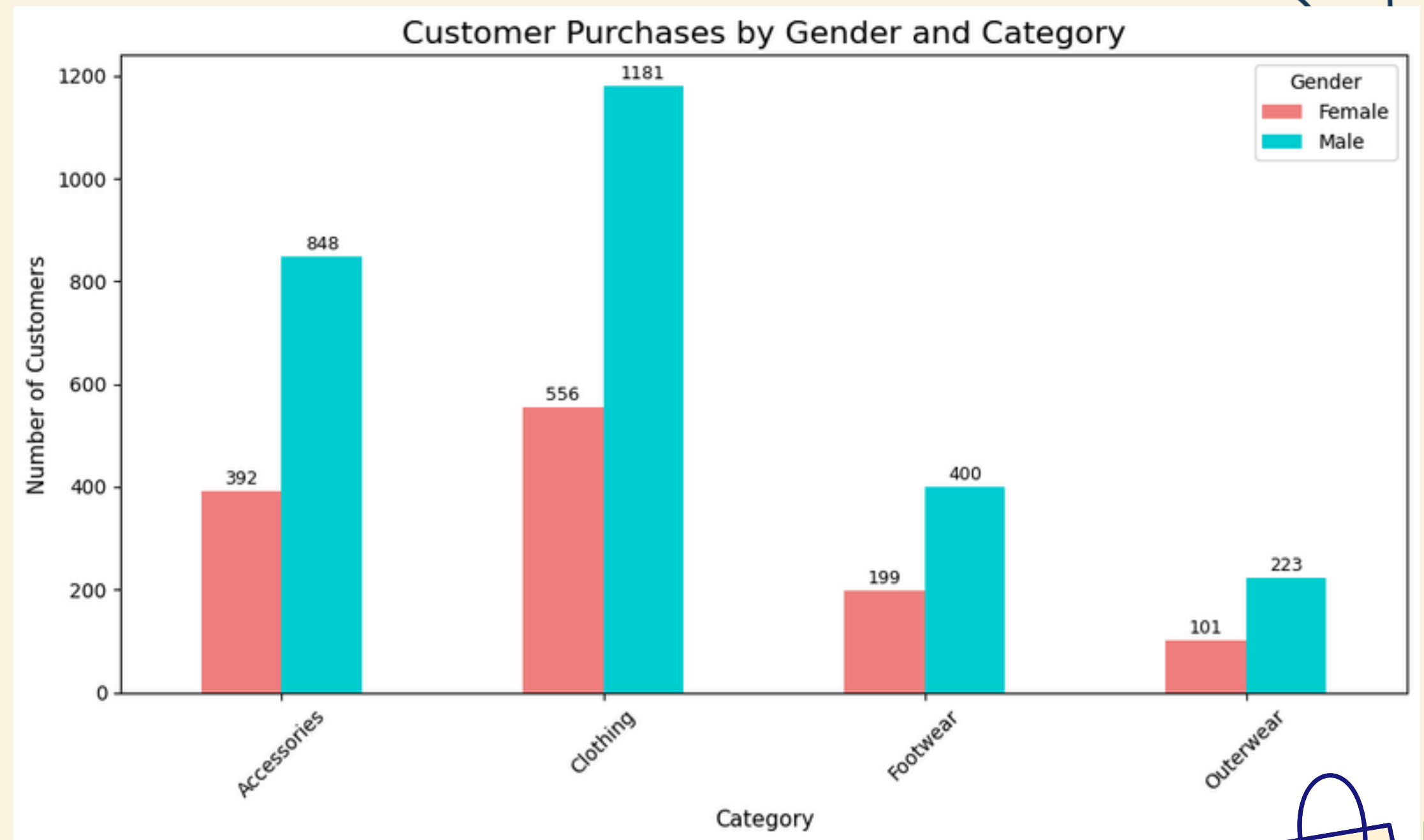




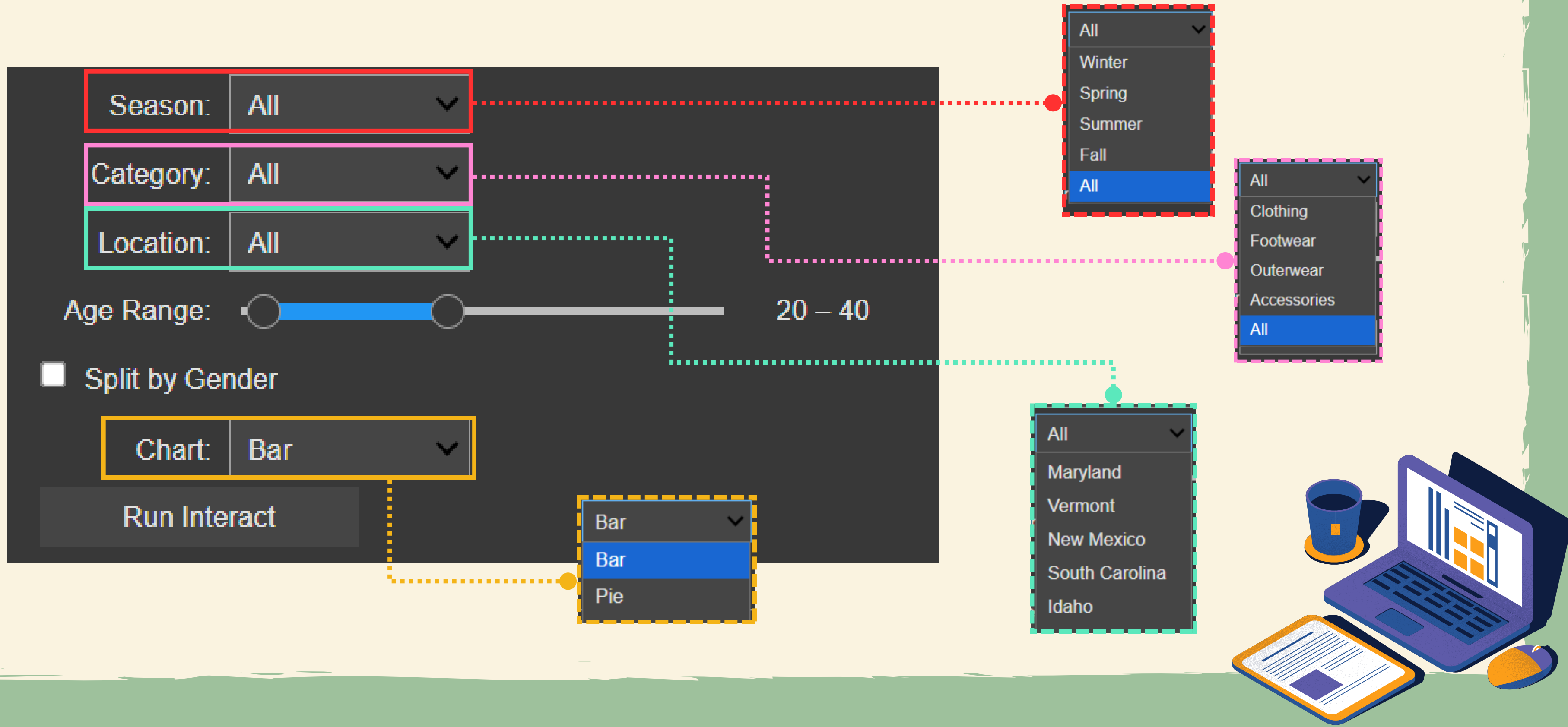
6. Customer Purchases by Gender and Category

Category	Gender	Count
Accessories	Female	392
	Male	848
Clothing	Female	556
	Male	1181
Footwear	Female	199
	Male	400
Outerwear	Female	101
	Male	223

- **Finding:** Male customers purchase Clothing & Accessories more than female customers.
- **Insight:** The primary target group is working-age men.



USER INTERFACE (UI)

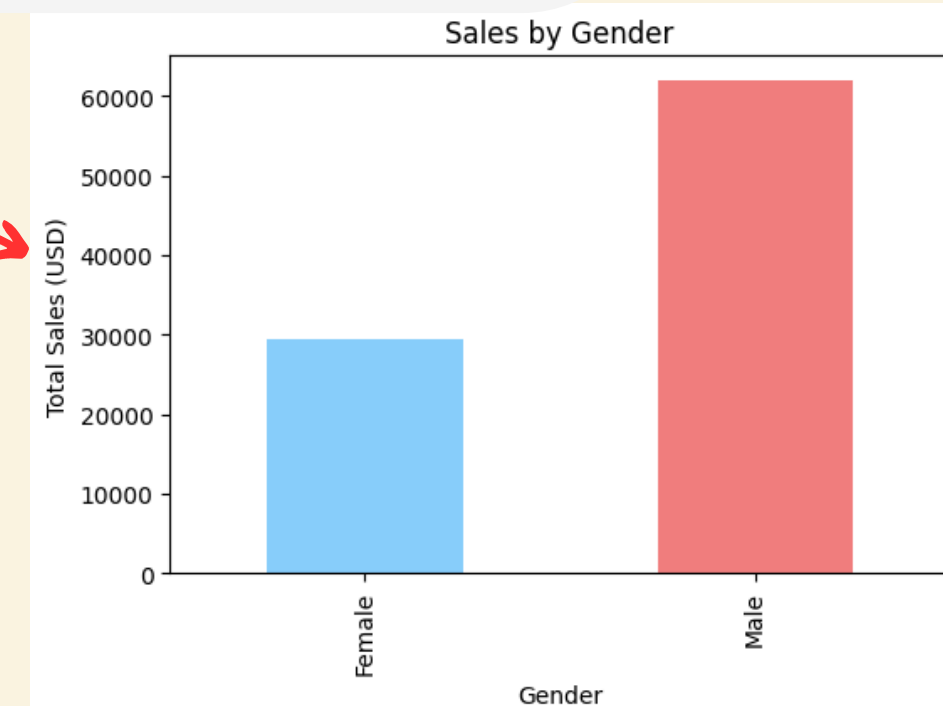


EX: USER INTERFACE (UI) 🔍

Season: All ▼
Category: All ▼
Location: All ▼
Age Range: 20 – 40
☒ Split by Gender
Chart: Bar ▼
Run Interact

Filtered Data Summary

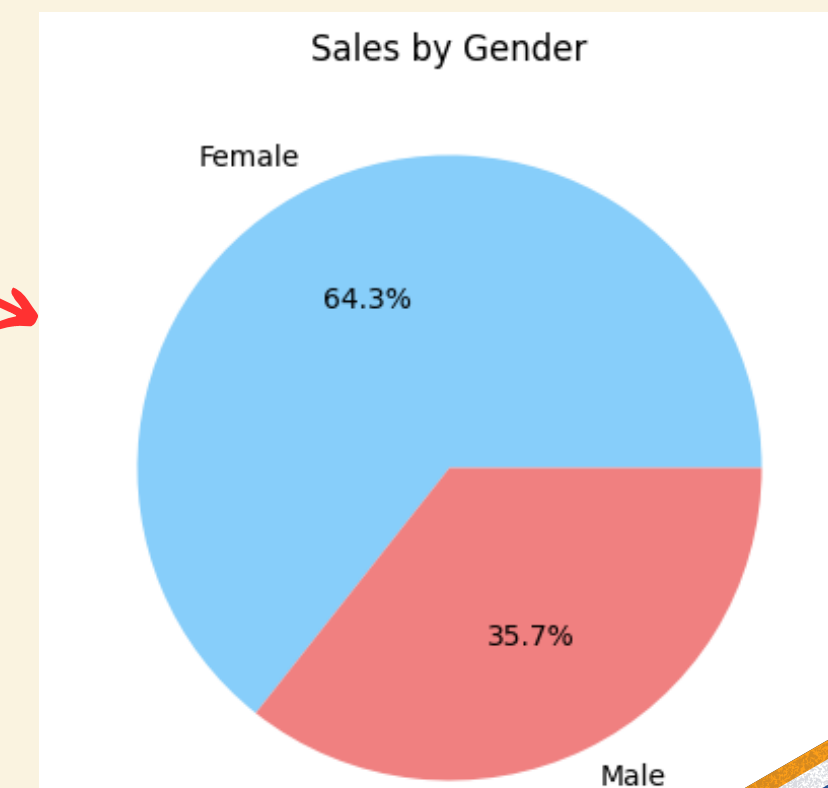
	Gender	Total Sales (USD)
0	Female	29426
1	Male	61966



Season: Spring ▼
Category: Footwear ▼
Location: Indiana ▼
Age Range: 20 – 70
☒ Split by Gender
Chart: Pie ▼
Run Interact

Filtered Data Summary

	Gender	Total Sales (USD)
0	Female	231
1	Male	128



THANK YOU

